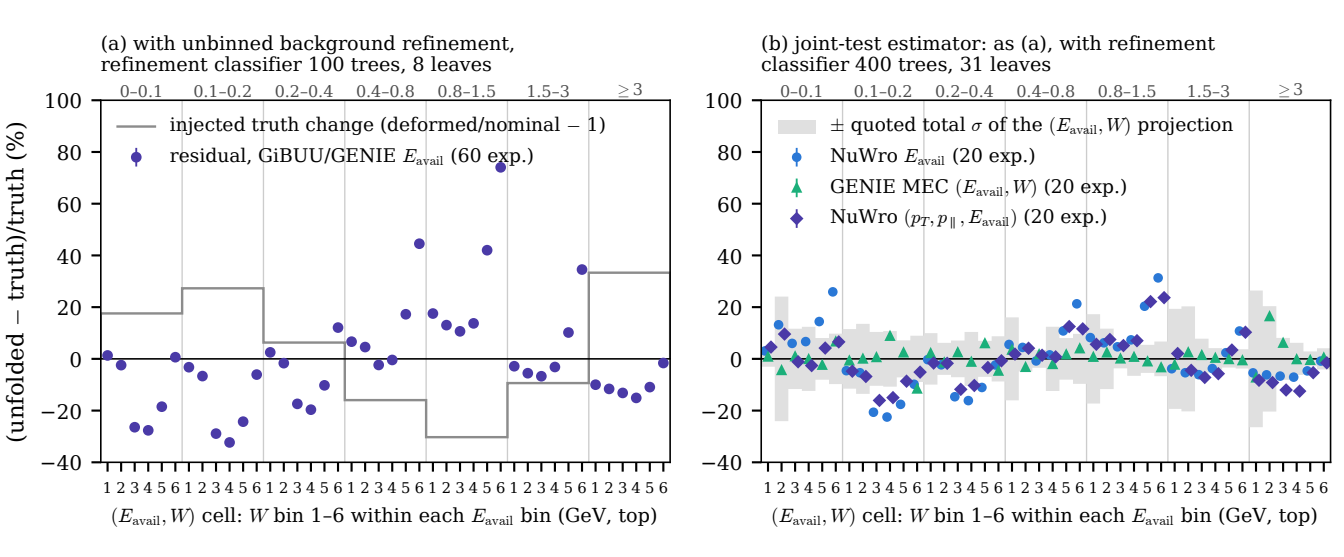
(a) with unbinned background refinement,
refinement classifier 100 trees, 8 leaves
0–0.1
0.1–0.2
0.2–0.4
0.4–0.8
0.8–1.5
1.5–3
≥3
injected truth change (deformed/nominal − 1)
residual, GiBUU/GENIE E_{avail} (60 exp.)
(unfolded − truth)/truth (%)
100
80
60
40
20
0
−20
−40
1 2 3 4 5 6
(E_{avail}, W) cell: W bin 1–6 within each E_{avail} bin (GeV, top)
(b) joint-test estimator: as (a), with refinement
classifier 400 trees, 31 leaves
± quoted total σ of the (E_{avail}, W) projection
NuWro E_{avail} (20 exp.)
GENIE MEC (E_{avail}, W) (20 exp.)
NuWro $(p_T, p_\parallel, E_{\text{avail}})$ (20 exp.)